

Finding 03 — Weak Telnet Credentials

Metasploitable 2 — Penetration Testing Finding Report

Target	Metasploitable 2
IP Address	192.168.116.129
Port	23/TCP
Service	Telnet
Finding	Weak / Default Credentials
Credentials observed	msfadmin : msfadmin
Result	Successful Telnet login and shell session
CVE	No specific CVE is assigned to the weak credentials themselves

1. Executive Summary

The target exposes Telnet on TCP port 23. During testing, the credentials **msfadmin:msfadmin** were accepted, resulting in a successful Telnet shell session. This demonstrates that the service is protected by weak/default credentials and that an attacker who discovers or guesses these credentials can gain interactive access to the host.

2. Evidence

The following screenshot shows the Metasploit Telnet login scanner reporting a successful login with the msfadmin account and an opened shell session.

```
msf6 auxiliary(scanner/telnet/telnet_login) > run
[*] 192.168.116.129:23 - No active DB -- Credential data will not be saved!
[*] 192.168.116.129:23 - 192.168.116.129:23 - Login Successful: msfadmin:msfadmin
[*] 192.168.116.129:23 - Attempting to start session 192.168.116.129:23 with msfadmin:msfadmin
[*] Command shell session 2 opened (192.168.116.128:44003 → 192.168.116.129:23) at 2026-09-01 17:50:45 +0300
[*] 192.168.116.129:23 - Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
msf6 auxiliary(scanner/telnet/telnet_login) > sessions

Active sessions
=====
Id  Name  Type  Information                                     Connection
--  --
2   shell TELNET msfadmin:msfadmin (192.168.116.129:23) 192.168.116.128:44003 → 192.168.116.129:23 (192.168.116.129)
msf6 auxiliary(scanner/telnet/telnet_login) > |
```

Observed output: **Login Successful: msfadmin** and **Command shell session opened**.

3. Technical Details

Port 23/TCP: Telnet is a remote interactive login protocol. In this test environment, the service permits authentication using the default/weak msfadmin credentials. Telnet also transmits authentication and session data without modern transport encryption, making it unsuitable for secure remote administration.

4. Impact

- Unauthorized users may obtain an interactive shell if they know or guess the weak credentials.
- Telnet traffic can expose credentials and session data because the protocol does not provide modern encryption.

- Successful access can provide a starting point for further enumeration and privilege escalation, depending on the account's privileges.

5. Remediation

- Disable Telnet if it is not required.
- Replace Telnet with SSH for remote administration.
- Immediately change/remove default credentials such as msfadmin:msfadmin.
- Use strong, unique passwords and avoid shared/default accounts.
- Restrict remote administration services with firewall rules or network segmentation.
- Monitor authentication logs for repeated failed or suspicious login attempts.

6. Risk Rating

Severity: High — In this lab, the weak/default credentials resulted in successful remote shell access. The exact severity in a real environment depends on account privileges, network exposure, and compensating controls.

7. Conclusion

The assessment confirmed a weak/default credential issue on the Telnet service running on port 23. The evidence demonstrates successful authentication and shell access using msfadmin:msfadmin. The recommended remediation is to disable Telnet, migrate to SSH, and remove default credentials.